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### Tree rooting enclosure apparatus

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#### Abstract

A tree rooting enclosure apparatus including a body housing with a cylindrical body base culminating into a pair of opposing end openings. The body and end openings are structured to accommodate a portion of a plant therethrough. The apparatus as a whole may be made of two removably engaged elongate halves structured to surround a selected portion of a plant and attach to itself forming a water-resistant fit around the selected portion of the plant.

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# Background/Summary

## FIELD OF THE INVENTION

(1) The present invention relates to systems and methods for propagating trees. In particular, the present invention relates to growing roots from an artificially created branch aperture enclosed in a container.

## BACKGROUND

(2) New trees can be generated from the branches of existing trees by manually creating a fissure or wound on a tree branch, which removes or injures a plant's phloem layer. The injury interrupts the flow of sugars and photosynthates to the roots, which then accumulate at the injury site. This accumulation when packed within a conducive medium generates root producing hormones. Therefore, enclosing the fissure with a wrapping filled with sphagnum moss or a potting mix facilitates the stimulation of adventitious root buds rather than leaves. Once a sufficient amount of time has passed, a mature network of roots may form to allow the branch to stand alone as a separate tree. The tree branch may then be removed from its original tree at a point below the newly formed network of roots and the removed branch may be planted and nurtured as a new tree.

(3) There exists a need for a rooting enclosure that efficiently and effectively envelops a manually created tree fissure when propagating trees. The enclosure should be optimally shaped to mimic the natural shape of the branch and should provide a number of fastening and stabilizing options for securing the enclosure that is resistant to water and animals. The enclosure should be reusable and modular in nature. Therefore, there exists a need in the art for a tree rooting enclosure apparatus and associated methods.

## SUMMARY OF THE INVENTION

(4) Embodiments of the present invention are related to a tree rooting enclosure apparatus including a body with a cylindrical body base, a pair of opposing end openings, and a plurality of spindles extending outward from the exterior edges of the pair of opposing end openings. The apparatus may be made of two removably engageable halves, each half being an identical duplicate of the other. Each half may be structured to mate with the other half when each respective half is inversely oriented with respect to the other half. Both halves when engaged may envelop at least a portion of a tree branch to propagate a rooting system.

(5) In this embodiment, the plurality of spindles may include at least one spindle shorter in length than the remaining spindles and the spindles may be structured in at least one of a symmetrically splayed orientation and an angled orientation. The cylindrical body base diameter may be proportional to the end opening diameters by a ratio of 1 to 0.75 respectively and the cylindrical body base may include a perimeter tab extending distally therefrom. The perimeter tab may include an elbow shaped T-lever structured to facilitate opening the apparatus when both halves are engaged. Furthermore, the perimeter tab may include four rounded corners and a pair of parallel sides and the pair of opposing end openings may form a concave void in the perimeter tab at each respective end of the apparatus.

(6) Another embodiment may include a tree rooting enclosure apparatus with a body including a cylindrical body base, a perimeter tab extending from the cylindrical body base, a pair of opposing end openings, a plurality of spindles extending outward from the exterior edges of the pair of opposing end openings, and a plurality of longitudinal channels extending the length of the cylindrical body base. The apparatus may be made of two removably engageable halves, each half being an identical duplicate of the other. Each half may be structured to mate with the other half when each respective half is inversely oriented with respect to the other half and both halves when engaged may envelop at least a portion of a tree branch to propagate a rooting system.

(7) In this embodiment, the plurality of spindles on each half may include at least one pair of longer

spindles flanking at least one shorter medial spindle. Additionally, the opposing sides of the perimeter tab may include a pair of fastening hooks structured to facilitate securing the tree rooting enclosure apparatus in a closed position with a perimeter fastening strap. Furthermore, at least one of the plurality of spindles may include an end distal to the end opening that is terminated by a cylinder. The perimeter tab of each half may include a concave protrusion structured to facilitate a user pulling apart the two halves when the two halves are engaged. The concave protrusion may be dimensioned to allow a user's finger to fit into the concave void and the perimeter tab may include a plurality of fastening apertures structured to accommodate a plurality of side cinches for securing the apparatus in a closed position.

(8) The perimeter tab may include at least one side with a plurality of joinder pegs and an opposing side with a plurality of joinder holes. The plurality of joinder pegs of the first half enclosing member may be structured to removably engage the plurality of joinder holes of the second half enclosing member to form a water-resistant fitting when the apparatus is in a closed position. The plurality of spindles may be in an angled orientation and may include at least one spindle shorter in length than the others. At least one spindle may include an end distal to the end opening that is terminated by a cylinder.

(9) Another embodiment may include a tree rooting enclosure apparatus with a body including a cylindrical body base body with a perimeter tab extending the entire longitudinal length of two sides of the cylindrical body base with at least one of a concave finger tab extension and an elbow shaped T-lever. It may include a pair of opposing end openings and a plurality of spindles extending outward from the exterior edges of the pair of opposing end openings. The apparatus may be made of two removably engageable halves, each half being an identical duplicate of the other and each half may be structured to mate with the other half when each respective half is inversely oriented with respect to the other half. Both halves when engaged may envelop at least a portion of a tree branch to propagate a rooting system. Furthermore, the concave finger tab protrusion and elbow shaped T-lever may be structured to facilitate a user pulling apart the two halves when the two halves are engaged.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top perspective view of a tree rooting enclosure apparatus according to an embodiment of the invention.

(2) FIG. 2 is a top perspective view of the tree rooting enclosure apparatus according to another embodiment of the invention.

(3) FIG. 3 is a top view of components of the tree rooting enclosure apparatus according to an embodiment of the invention.

(4) FIG. 4 is a side perspective view of components of the tree rooting enclosure apparatus being fitted together according to an embodiment of the invention.

(5) FIG. 5A is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

(6) FIG. 5B is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

(7) FIG. 6A is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

(8) FIG. 6B is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

(9) FIG. 6C is a top perspective view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.

- (10) FIG. 7A is a lateral end view of the tree rooting enclosure apparatus according to another embodiment of the invention.
- (11) FIG. 7B is a segmented view comparing different end embodiments of the tree rooting enclosure apparatus.
- (12) FIG. 7C is a segmented environmental perspective view of one end of the tree rooting enclosure apparatus according to an embodiment of the invention.
- (13) FIG. 8. is a perspective environmental view of a tree rooting enclosure apparatus in an opened position according to an embodiment of the invention.
- (14) FIG. 9 is a top perspective environmental view of a tree rooting enclosure apparatus in a closed position according to an embodiment of the invention.
- (15) FIG. 10 is a side perspective environmental view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.
- (16) FIG. 11 is a top perspective environmental view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.
- (17) FIG. 12 is a side perspective environmental view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.
- (18) FIG. 13 is a cross sectional environmental view of the tree rooting enclosure apparatus according to an embodiment of the invention.
- (19) FIG. 14A is an environmental view of a root network mass created by the tree rooting enclosure apparatus according to an embodiment of the invention.
- (20) FIG. 14B is an environmental view of a root network mass created by the tree rooting enclosure apparatus according to an embodiment of the invention.
- (21) FIG. 15A is a side environmental view of a root network mass created by the tree rooting enclosure apparatus with a stabilizing stake fitted thereto.
- (22) FIG. 15B is a front environmental view of a root network mass created by the tree rooting enclosure apparatus with a pair of stabilizing stakes fitted thereto.
- (23) FIG. 16A illustrates a different embodiment of the tree rooting enclosure apparatus.
- (24) FIG. 16B illustrates a different embodiment of the tree rooting enclosure apparatus.
- (25) FIG. 16C illustrates a different embodiment of the tree rooting enclosure apparatus.
- (26) FIG. 16D illustrates a different embodiment of the tree rooting enclosure apparatus.
- (27) FIG. 16E illustrates a different embodiment of the tree rooting enclosure apparatus.
- (28) FIG. 16F illustrates a different embodiment of the tree rooting enclosure apparatus.
- (29) FIG. 17 illustrates another embodiment of the tree rooting enclosure apparatus.
- (30) FIG. 18 is a perspective view of one half of the tree rooting enclosure apparatus according to an embodiment of the invention.
- (31) FIG. 19 is a top view of one half of the tree rooting enclosure apparatus according to the embodiment of the invention illustrated in FIG. 18.
- (32) FIG. 20A is a front view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.
- (33) FIG. 20B is a perspective view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.
- (34) FIG. 20C is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.
- (35) FIG. 20D is a top view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.
- (36) FIG. 21 is a perspective view of one half of the tree rooting enclosure apparatus according to an embodiment of the invention.
- (37) FIG. 22 is a top view of one half of the tree rooting enclosure apparatus according to the embodiment of the invention illustrated in FIG. 21.
- (38) FIG. 23A is a front view of componentry of the tree rooting enclosure apparatus illustrated in

FIG. 21.

(39) FIG. 23B is a perspective view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

(40) FIG. 23C is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

(41) FIG. 23D is a top view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

(42) FIG. 24 is a perspective view of one half of the tree rooting enclosure apparatus according to an embodiment of the invention.

(43) FIG. 25A is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 24.

(44) FIG. 25B is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 24.

(45) FIG. 25C is a front view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 24.

#### DETAILED DESCRIPTION OF THE INVENTION

(46) The present invention will now be described in detail with reference to the accompanying drawings. The embodiment descriptions are illustrative and not intended to be limiting in any way. Other embodiments of the invention will readily suggest themselves to persons with ordinary skill in the art after having the benefit of this disclosure. Accordingly, the following embodiments are set forth without any loss of generality and without imposing limitation upon the claimed invention.

(47) Directional terms such as “above” “below” “upper” “lower” and other like terms are used for the convenience of the reader in reference to the drawings. Additionally, the description may contain terminology to convey position, orientation, and direction without departing from the principles of the present invention. Such positional language should be taken in context of the represented drawings.

(48) Quantitative terms such as “generally” “substantially” “mostly” and other like terms are used to mean that the referred object, characteristic, or quality constitutes a majority of the referenced subject. Likewise, use of the terms such as first and second do not necessarily designate a limitation of quantity. Such terms may be used as a method of describing the presence of at least one of the referenced elements or may provide a means of differentiating orientation. The meaning of any term within this description is dependent upon the context within which it is used, and the meaning may be expressly modified.

(49) Referring now to FIG. 1, the tree rooting enclosure apparatus **100**, hereinafter the apparatus **100**, is shown. The apparatus **100** may include an apparatus first end **101**, an apparatus second end **102**, a body **150** with cylindrical body base **103** and a pair of body tapers **106** at a body first end **121** and a body second end **122**. The apparatus **100** may also include what may be designated as an apparatus right side **111** and an apparatus left side **112**.

(50) The body tapers **106** may be a pair of opposing tapered ends culminating into a pair of opposing end openings **104**, **105**. As such, there may be a first end opening **104** and a second end opening **105**. In some embodiments, the opposing end openings **104**, **105** may include a diameter relative to the diameter of the cylindrical body base **103**. By way of non-limiting example, this may be a cylindrical body base diameter to end opening diameter ratio of 2.5 to 1 respectively.

Furthermore, extending distally from the edges of the end openings **104**, **105** may be a plurality of spindles **108**. In this particular embodiment, the spindles **108** are shown having equivalent length. Additionally, in some embodiments, the spindles **108** may include spindle ends **109** that may be bulbous.

(51) The apparatus **100** may also include a perimeter tab **107** that may extend distally and surround the cylindrical body base **103** and body tapers **106** along a longitudinal axis of the apparatus **100**. In some embodiments the perimeter tab **107** may be polygonal and structured as an octagonal lip

with rounded vertices. In fact, as will be described more fully hereafter, the apparatus **100** may be constructed from two congruent enclosing members each comprising a longitudinal half of the apparatus **100** when bifurcated along its longitudinal medial axis. Therefore, as will be described, the apparatus **100** as a whole is structured from two removably engaged halves forming a water-resistant fit. It should be noted that the water-resistant fit also protects against intrusion from animals such as birds, squirrels, and the like.

(52) The perimeter tab **107** may extend around the perimeter of each enclosing member. As such, the perimeter tab **107** may include a plurality of joinder pegs **115** extending distally from the interior right underside of the perimeter tab **107**. Furthermore, the exterior of the perimeter tab **107** may also include a plurality of fastening dowels **110** protruding therefrom. In this embodiment, there are two fastening dowels **110** asymmetrically positioned in a staggered formation on the apparatus right side **111** and apparatus left side **112**. Additionally, the perimeter tab **107** may include a plurality of fastening apertures **114**, which in some embodiments may be slits or holes for tying the apparatus **100** together at its midsection.

(53) FIG. 2 illustrates that in some embodiments the apparatus **100** may include a plurality of longitudinal channels **201** that may extend the length of the cylindrical body base **103**. A longitudinal channel **201** may be a concave indentation in the exterior of the cylindrical body base **103**. In this particular embodiment, there may be two longitudinal channels **201** on opposing surfaces of the apparatus **100**. One skilled in the art will appreciate that there may be more or less longitudinal channels **201** within the apparatus **100** depending on need, preference, and circumstance. Likewise, in some embodiments, the longitudinal channels **201** may extend past the cylindrical body base **103** into the body taper **106** and in some embodiments may create an indented arch therein.

(54) FIG. 3 illustrates that the apparatus **100** may be made of two congruent enclosing members **303a**, **303b** structured to removably engage one another. Further shown are a plurality of joinder holes **301** on the apparatus left side **112** sized and positioned to accommodate the plurality of joinder pegs **115** on the apparatus right underside **111**. As such, the perimeter tab **107** may include at least one side with a plurality of joinder pegs **115** and an opposing side with a plurality of joinder holes **301**. In some embodiments, the plurality of joinder pegs **115** may include an exterior **302** with a facilitating groove to assist with positioning and fitting the joinder pegs **115** into and out of the joinder holes **301** with a friction fit. Furthermore, the opposing end openings **104**, **105** may be formed from semi-circular medial openings **304** at opposing ends of each enclosing member **303a**, **303b**.

(55) FIG. 4 shows how the two half enclosing members **303a**, **303b** may be removably engaged. As shown, each enclosing member **303a**, **303b** is an identical duplicate of the other. A first enclosing member **303a** is positioned in an inverse orientation to a second enclosing member **303b**. As such, the plurality of joinder pegs **115** on the right side **111** of the first enclosing member **303a** are aligned with the plurality of joinder holes **301** on the left side **112** of the second enclosing member **303b**. A user may removably engage the first and second enclosing members **303a**, **303b** by exerting force **401** to combine the joinder pegs **115** and joinder holes **301** with a friction fit thereby placing the apparatus **100** in a closed position.

(56) FIGS. 5A and 5B are lateral end views of the apparatus **100**. As shown, the joinder pegs **115** are aligned with the joinder holes **301** and the enclosing member halves **303a**, **303b** are stacked vertically. FIG. 5B shows that when fitted together, the two enclosing member halves **303a**, **303b** create a water-resistant fit **502** at the perimeter tab **107** when the apparatus **100** is in a closed position. Also shown is the comparison between the cylindrical body base diameter **504** and an end opening diameter **503**. As such, by way of non-limiting example, the ratio between the cylindrical body base diameter **504** at its widest point and an end opening diameter **503** may be 2.5 to 1 respectively in the closed position. However, one skilled in the art will appreciate that the ratio may be slightly larger or slightly smaller depending on need, preference, and circumstance.

(57) FIGS. 6A and 6B are lateral end views of the apparatus **100** comparing two slightly different embodiments, but with similar spindle **108** configurations. In one embodiment, as shown in FIG. 6A, the apparatus **100** may include a smooth body base **601**. Meaning, this embodiment does not include the longitudinal channel **201** undulating the body base **103**. However, the ratio between the cylindrical body base diameter **504** at its widest point and an end opening diameter **503** may still be 2.5 to 1 respectively. But, similar to the previous embodiment, one skilled in the art will appreciate that the ratio may be larger or smaller depending on need, preference, and circumstance. In fact, the entire apparatus **100** may be made larger or smaller to accommodate differently sized plants and branches. By way of non-limiting example, the apparatus **100** as a whole may be much smaller making the end openings **104, 105** much smaller. Likewise, the apparatus **100** may be made much larger to accommodate larger branches and plants whereby the end openings **104, 105** and cylindrical body base diameter **504** would be much larger. Likewise, one skilled in the art will appreciate that the ratio between the end opening diameter **503** and the cylindrical body base diameter **504** may be larger or smaller depending on need, preference, and circumstance.

(58) FIG. 6B illustrates by way of comparison against FIG. 6A that in some embodiments the apparatus **100** may include a longitudinal channel **201** thereby undulating the body base **103**. However, both FIGS. 6A and 6B show an embodiment whereby the spindles **108** are in a symmetrically splayed configuration **602**. This means that the spindles **108** are level with, and evenly distributed around the edge of the end opening **104**. In this embodiment the spindles **108** are also of the same length.

(59) FIG. 6C illustrates how the apparatus **100** may be secured when in a closed position around a tree branch **604** with symmetrically splayed spindles **602**. As shown, the spindles **108** may fit around the tree branch **604** at each end opening **104, 105** and may be tightened therearound by at least one end cinch **603** at each. By tightening the end cinch **603** around the spindles **108**, the apparatus **100** in turn is more firmly secured to a tree branch **604** at a desired location. However, one skilled in the art will appreciate that the end cinch **603** may or may not be used to supplement securement of the apparatus **100** depending on need, preference, and circumstance.

(60) FIG. 7A is a lateral end view of the apparatus **100** showing an embodiment with angled spindles **701**. In this embodiment, the angled spindles **701** are biased toward the center of the end openings **104, 105** creating a more fitted mount on a tree branch **604**. Here, the angled spindles **701** narrow the end openings **104, 105** so that when a tree branch **606** is fitted therethrough, the angled spindles **701** are already secured thereto without the need for end cinches **603**.

(61) FIG. 7B illustrates a comparison between the symmetrically splayed spindles **602** and the angled spindles **701**. Here, the spindle angle **705** is clearly shown as biasing toward the center of the end opening **104**. Furthermore, the angled spindles **701** may be staggered or alternating in their length **704**. Meaning, larger spindles **702** may be alternated with smaller spindles **703**. By way of non-limiting example, the larger spindles **702** may all be the same length and the smaller spindles **703** may all be the same length. Furthermore, they may all share the same biasing angle toward the center.

(62) FIG. 7C is a segmented environmental view of a first end **104** of the apparatus **100** with angled spindles **701**. As shown, the tree branch **604** is clasped by the angled spindles **701** resulting in the apparatus **100** being secured in place. By way of non-limiting example, the angled spindles **701** may be bendable and not rigid in order to create a more custom and secure fit.

(63) FIG. 8 begins to illustrate some of the functional features of the apparatus **100**. As shown, the apparatus **100** is in an opened position **800**. A tree branch **604** with a manually created fissure **803** is positioned along a medial longitudinal axis of the apparatus **100** between the first and second enclosing members **303a, 303b**.

(64) Exposed by the open position **800** is the apparatus body internal cavity **801** created by the cylindrical body base **103** and the body tapers **106** that house the tree branch **604**. Also shown is an interior longitudinal convex ridge **802** formed by the longitudinal channel **201** on the exterior of the



apparatus **100**. Hence, each longitudinal channel **201** of the plurality of longitudinal channels **201** includes a correlating interior longitudinal convex ridge **802**. As will be explained more fully hereafter, the plurality of interior longitudinal convex ridges **802** may be structured to create channel indentions within a root network mass formed within the internal cavity **801** of the apparatus **100**.

(65) FIG. **9** illustrates the apparatus **100** in a closed position **900** with a tree branch **604** positioned along the medial longitudinal axis of the apparatus **100** between the first and second enclosing members **303a**, **303b** within the internal cavity **801**. The tree branch **604** is protruding through the first and second end openings **104**, **105** while the apparatus **100** is secured shut by the plurality of joinder pegs **115** engaging the plurality of joinder holes **301**. Therefore, the first half enclosing member **303a** and second half enclosing member **303b** are shown as being structured to removably engage each other around a portion of a plant in a closed position **900** to form the whole apparatus **100**.

(66) FIGS. **10** and **11** illustrate how the apparatus **100** may further be secured around a tree branch **604** utilizing the fastening dowels **110**. As shown, when in the closed position **900** a perimeter fastening strap **1001**, which by way of non-limiting example may be a rubber band, may be secured around the fastening dowel **110** extending from the right side **111** of a first enclosing member **303a**, wrapped around the apparatus **100** and secured to a fastening dowel **110** positioned on an inverted second enclosing member **303b**. Therefore, the perimeter fastening strap **1001** is able to be wrapped around the exterior of the apparatus **100** further securing it in the closed position **900** around a tree branch **604**.

(67) FIG. **12** illustrates another securing means by which the apparatus **100** may be fastened in the closed position **900** around a tree branch **604**. As shown, when in the closed position **900** the fastening apertures **114** located on the perimeter tab **107** may align. This alignment may facilitate the apparatus **100** being further secured closed around a tree branch **604** when a plurality of side cinches **1201** are threaded through the aligned fastening apertures **114**. In some embodiments, the side cinches **1201** may be a standard cable tie.

(68) FIG. **13** is a cross sectional environmental view of the apparatus **100** illustrating the newly formed roots **1301** that are created within the internal cavity **801**. As shown, the tree branch **604** may be enclosed within the apparatus **100** and may extend through the first and second end openings **104**, **105**. After the manually created fissure **803** has been created, it may be enclosed with a potting mix **1302** such as peat moss, sphagnum moss, or the like. The tree branch **604** then begins to create the newly formed roots **1301**, which ultimately results in a root network mass **1401**.

(69) FIGS. **14A** and **14B** are environmental views of the root network mass **1401** created by the apparatus **100**. This root network mass **1401** is formed by a combination of a mature network of newly formed roots **1301** and either a sphagnum moss or potting mix **1302** that have been compacted by the apparatus **100**. This root network mass **1401** is dense enough and compacted tightly enough within the apparatus **100** to have the plurality of interior longitudinal convex ridges **802** form a plurality of longitudinal indentions **1402** within the root network mass **1401**.

(70) FIG. **14B** shows that the tree branch may be severed **1404** proximate one open end **105** so that it may be planted. It also illustrates that the longitudinal indentions **1402** may be on opposing sides of the root network mass **1401**.

(71) FIGS. **15A** and **15B** are environmental views of the root network mass **1401** and planted branch **1500** in its own separate soil **1501**. They further show that the plurality of longitudinal indentions **1402** may be used as tracks to leverage stakes **1502** aligned therewith to supply support for the root network mass **1401** and base of the planted branch **1500**. This may allow for the planted branch to grow upright and mature in a healthy uniform fashion. Similar to FIG. **14B**, FIG. **15B** illustrates that in some embodiments two stakes **1502** may align with the longitudinal indentions **1402** on opposing sides of the root network mass **1401**.

(72) FIGS. **16A** through **16F** illustrate different embodiments of the apparatus **100** that are contemplated to be within the scope of this application. As such, FIG. **16A** shows an apparatus without body tapers **106**. In this embodiment, open ends **104**, **105** may be medially positioned within the cylindrical body base **103**. FIG. **16B** illustrates that in some embodiments, an apparatus first end **101** may include a single body taper **1602** and an apparatus second end **102** may include only the truncated end **1603** of a cylindrical body base **103**. FIG. **16C** may be the same as the embodiment shown in FIG. **16A** with the addition of at least one longitudinal channel **201** positioned thereon. FIG. **16D** illustrates an embodiment whereby the cylindrical body base **103** includes inverted tapered ends **1605**. In this embodiment, the open ends **104**, **105** may be medially positioned between the inverted tapered ends **1602** within the cylindrical body base **103**. FIG. **16E** is similar to FIG. **16D**, however, it includes rounded vertices at the inverted tapered ends **1605**. Additionally, FIG. **16F** includes an embodiment whereby one end of the cylindrical body base **103** includes inverted tapered ends **1605** and the other includes a truncated end **1603**.

(73) FIG. **17** illustrates another embodiment of the invention whereby the longitudinal channel **201** is supplemented with a plurality of lateral channels **1701**. The plurality of lateral channels **1701** may form a plurality of interior lateral convex ridges structured to create lateral channel indentions within a root network mass **1401** formed inside the apparatus **100**. These lateral channel indentions may assist with stabilizing and leveraging the root network mass **1401** when it is planted in its own separate soil **1501**. In other words, these lateral channel indentions within the root network mass **1401** may fill in with dirt when planted and may stabilize the planted branch **1500** as well as prevent it from easily being pulled out of its own separate soil **1501**.

(74) FIGS. **18** and **19** illustrate another embodiment of the invention whereby the perimeter tab **107** may include elbow shaped T-levers **1804** structured to facilitate opening the apparatus **100** when both halves are engaged. Furthermore, this embodiment illustrates that the plurality of spindles **108** may include at least one spindle **108** shorter in length than the others **703** and at least one spindle end **109** distal to the end opening **1801** that is terminated by a cylinder **1805**. The spindles may also be angled **705** and in some embodiments, the plurality of spindles **108** on each half may include at least one pair of longer spindles **702** flanking at least one shorter medial spindle **703**.

(75) Additionally, the perimeter tab **107** may include four rounded corners **1803** and a pair of parallel sides **1802**. Each side **1802** of each half may be completely surrounded or bordered by the perimeter tab **107**. At each longitudinal end **1801** of each half of the apparatus **100**, the perimeter tab **107** may converge into the end opening **1801**. In other words, the opposing end openings may form a concave void in the perimeter tab **107** at each respective end of the apparatus **100**.

Furthermore, in some embodiments, the width of the body **150** compared to the end opening **1801** may equal 1 to 0.75 respectively. In some embodiments, the plurality of spindles **108** may include five (5) spindles **108** per half, per end. In other embodiments, the plurality of spindles **108** may include two (2) spindles **108** per half, per end. However, one skilled in the art will appreciate that the number of spindles **108** may be more or less depending on need, preference, and circumstance.

(76) FIGS. **20A** through **20D** show the elbow shaped T-levers **1804** in more detail. As shown, they may include a pair of opposing T-flanks **2001** on either side of a drop member **2004**. The drop member **2004** may be perpendicular to an extender **2003** that protrudes from the perimeter tab **107** forming the elbow **2002**.

(77) FIGS. **21** and **22** illustrates another embodiment of the invention whereby the perimeter tab **107** includes a finger tab **2101** structured to facilitate a user opening the apparatus **100** when both halves are engaged. The finger tab **2101** may extend outward from the perimeter tab **107** and may include a u-shaped member designed to fit a standard user's finger within the u-shaped void.

(78) FIGS. **23A** through **23D** illustrate the finger tab **2101** in more detail. As shown, the finger tab **2101** may include a pair of extending arms **2302** connected by a concave pull **2301**. The extending arms **2302** may cause the finger tab **2101** to extend distally from the perimeter tab **107** and the concave pull **2301** may be sized and dimensioned to allow the concave finger tab protrusion **2101**

to facilitate a user's finger fitting into the created concave void. The concave pull **2301** may be used by a user to leverage and pull the perimeter tab **107** and apparatus half away from its engaged other half.

(79) FIG. **24** illustrates another embodiment of the invention whereby opposing sides **1801** of the perimeter tab **107** may include a pair of fastening hooks **2401** structured to facilitate securing the apparatus **100** in a closed position with a perimeter fastening strap. The fastening hooks **2401** may abut an aperture **2403** within the perimeter tab **107**. This figure also emphasizes that in some embodiments there may be as few as two spindles **2402**. It also highlights that the spindle ends **109** may be terminated by cylinders **1805**.

(80) FIGS. **25A** through **25C** emphasize the fastening hooks **2401** in more detail. As shown, they may be positioned on the perimeter tab **107** and may include a hook housing **2501**, a hook hood **2502**, and a hook void **2503** within the hook housing. The hook hood **2502** may allow for a fastening strap to remain in place without prolapsing or slipping off entirely.

## Claims

1. A tree rooting enclosure apparatus comprising: a body comprising: a cylindrical body base, a pair of opposing end openings with edges exterior to the body; and a plurality of spindles longitudinally oriented with respect to the cylindrical body base and extending outward from the exterior edges of the pair of opposing end openings; wherein the tree rooting enclosure apparatus is comprised of two removably engageable halves, each half being an identical duplicate of the other; wherein each half is configured to mate with the other half when each respective half is inversely oriented with respect to the other half; and wherein both halves when engaged envelop at least a portion of a tree branch to propagate a rooting system.
2. The tree rooting enclosure apparatus of claim 1 wherein the plurality of spindles comprises at least one spindle shorter in length than the remaining spindles.
3. The tree rooting enclosure apparatus of claim 1 wherein the spindles are configured in at least one of a symmetrically splayed orientation and an angled orientation.
4. The tree rooting enclosure apparatus of claim 1 wherein the cylindrical body base diameter is proportional to the end opening diameters by a ratio of 1 to 0.75 respectively.
5. The tree rooting enclosure apparatus of claim 1 wherein the cylindrical body base comprises a perimeter tab extending distally therefrom.
6. The tree rooting enclosure apparatus of claim 5 wherein the perimeter tab comprises an elbow shaped T-lever configured to facilitate opening the tree rooting enclosure apparatus when both halves are engaged.
7. The tree rooting enclosure apparatus of claim 5 wherein the perimeter tab comprises four rounded corners and a pair of parallel sides.
8. The tree rooting enclosure apparatus of claim 5 wherein the pair of opposing end openings form a concave void in the perimeter tab at each respective end of the tree rooting enclosure apparatus.
9. A tree rooting enclosure apparatus comprising: a body comprising: a cylindrical body base, a perimeter tab extending from the cylindrical body base, and a pair of opposing end openings with edges exterior to the body; a plurality of spindles longitudinally oriented with respect to the cylindrical body base and extending outward from the exterior edges of the pair of opposing end openings; and a plurality of longitudinal channels extending the length of the cylindrical body base; wherein the tree rooting enclosure apparatus is comprised of two removably engageable halves, each half being an identical duplicate of the other; wherein each half is configured to mate with the other half when each respective half is inversely oriented with respect to the other half; and wherein both halves when engaged envelop at least a portion of a tree branch to propagate a rooting system.
10. The tree rooting enclosure apparatus of claim 9 wherein the plurality of spindles comprises at

least one pair of longer spindles flanking at least one shorter medial spindle.

11. The tree rooting enclosure apparatus of claim 9 wherein opposing sides of the perimeter tab comprise a pair of fastening hooks configured to facilitate securing the tree rooting enclosure apparatus in a closed position with a perimeter fastening strap.

12. The tree rooting enclosure apparatus of claim 9 wherein at least one of the plurality of spindles comprises an end distal to its respective end opening that is terminated by a cylinder.

13. The tree rooting enclosure apparatus of claim 9 wherein the perimeter tab of each half includes a concave protrusion configured to facilitate a user pulling apart the two halves when the two halves are engaged.

14. The tree rooting enclosure apparatus of claim 13 wherein the concave protrusion is dimensioned to allow a user's finger to fit into its concave void.

15. The tree rooting enclosure apparatus of claim 9 wherein the perimeter tab comprises a plurality of fastening apertures configured to accommodate a plurality of side cinches for securing the tree rooting enclosure apparatus in a closed position.

16. The tree rooting enclosure apparatus of claim 9 wherein the perimeter tab comprises at least one side with a plurality of joinder pegs and an opposing side with a plurality of joinder holes.

17. The tree rooting enclosure apparatus of claim 16 wherein the plurality of joinder pegs of the first half enclosing member is configured to removably engage the plurality of joinder holes of the second half enclosing member to form a water-resistant fitting when the apparatus is in a closed position.

18. The tree rooting enclosure apparatus of claim 9 wherein the spindles are in an angled orientation.

19. The tree rooting enclosure apparatus of claim 9 wherein the plurality of spindles includes at least one spindle shorter in length than the others and at least one spindle comprises an end distal to its respective end opening that is terminated by a cylinder.

20. A tree rooting enclosure apparatus comprising: a body comprising: a cylindrical body base, a perimeter tab extending the entire longitudinal length of two sides of the cylindrical body base comprising at least one of a concave finger tab extension and an elbow shaped T-lever, a pair of opposing end openings with edges exterior to the body; a plurality of spindles longitudinally oriented with respect to the cylindrical body base and extending outward from the exterior edges of the pair of opposing end openings; wherein the tree rooting enclosure apparatus is comprised of two removably engageable halves, each half being an identical duplicate of the other; wherein each half is configured to mate with the other half when each respective half is inversely oriented with respect to the other half; wherein both halves when engaged envelop at least a portion of a tree branch to propagate a rooting system; and wherein the concave finger tab protrusion and elbow shaped T-lever are configured to facilitate a user pulling apart the two halves when the two halves are engaged.

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